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Department of Computer Science and Engineering

Ingredient Analysis System for Personalized Dietary Restrictions

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Abstract

Food ingredient analysis has become increasingly important because many individuals need to avoid specific ingredients due to allergies, medical conditions, or lifestyle preferences. However, manually inspecting food labels is often challenging because ingredient lists may be lengthy, unclear, or difficult to read from captured images.

This project presents **LabelLens**, an intelligent ingredient analysis system designed to evaluate food labels according to selected dietary restrictions. The system accepts either food-label images or manually entered ingredient text and provides an explainable dietary assessment.

The proposed system combines image preprocessing, Optical Character Recognition (OCR), text normalization, deep learning classification, and rule-based dietary restriction analysis. OpenCV is used for image preprocessing, while EasyOCR is applied to extract ingredient information from food-label images. The extracted or manually provided ingredient text is cleaned and prepared before being processed by trained deep learning models.

Three deep learning models, namely Long Short-Term Memory (LSTM), Convolutional Neural Network (CNN), and Deep Neural Network (DNN), were developed and evaluated using the prepared ingredient dataset. Based on experimental results, CNN and DNN achieved the highest classification accuracy. However, the DNN model produced a lower validation loss and demonstrated better suitability for structured ingredient-text classification. Therefore, the DNN model was selected as the final prediction model and integrated into the application.

The final system is implemented using Django and provides a user-friendly web interface where users can upload food labels or enter ingredient information. The prediction output is combined with an explainable restriction matching system to identify potentially problematic ingredients and generate personalized recommendations.

The developed system demonstrates that deep learning combined with explainable analysis techniques can provide an effective and practical solution for automated dietary restriction assessment.

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Chapter 1

Introduction

1.1 Background

Food consumption decisions have become increasingly complex due to the growing number of dietary requirements among individuals. Many people need to avoid certain ingredients because of food allergies, intolerances, medical conditions, or personal lifestyle choices. Common dietary restrictions include lactose intolerance, gluten sensitivity, nut allergies, sugar monitoring, and vegan preferences.

Although packaged food products contain ingredient information, manually interpreting these labels can be difficult. Ingredient names may appear in different forms, some components may be hidden under technical names, and the text may become difficult to read when captured through a camera. These challenges increase the possibility of incorrect dietary decisions.

With the advancement of Artificial Intelligence (AI), computer vision, and deep learning technologies, automated food-label analysis systems have become a promising solution. Such systems can extract information from food labels, analyze ingredient content, and provide users with meaningful recommendations.

This project introduces **LabelLens**, a deep learning-based ingredient analysis system that combines OCR technology, neural network classification, and explainable rule-based assessment to help users understand whether a food product satisfies their selected dietary requirements.

1.2 Problem Statement

Many individuals face difficulties when determining whether packaged food products are suitable for their dietary restrictions. Traditional manual inspection requires users to carefully examine ingredient lists, which can be time-consuming and error-prone.

Furthermore, food labels captured through mobile cameras often contain challenges such as poor lighting, image distortion, small text size, and background noise. These problems can reduce the accuracy of ingredient extraction.

Existing automated solutions may provide predictions without explaining which ingredients caused a particular decision. For dietary applications, transparency is important because users need to understand the reason behind a recommendation.

Therefore, this project aims to develop an intelligent system that can extract ingredient information, classify food content using deep learning techniques, and provide an explainable dietary restriction assessment.

1.3 Objectives

The main objective of this project is to develop a deep learning-based web application capable of analyzing food ingredients and providing personalized dietary restriction recommendations.

The specific objectives are:

- To develop a Django-based web application for automated ingredient analysis.
- To extract ingredient information from food-label images using OCR techniques.
- To preprocess and normalize ingredient text for deep learning classification.
- To develop and compare LSTM, CNN, and DNN-based classification models.
- To evaluate deep learning models using accuracy, precision, recall, F1-score, and validation loss.
- To select the most suitable deep learning model for deployment.
- To integrate the selected model with an explainable dietary restriction analysis module.
- To provide users with understandable dietary recommendations based on detected ingredients.

1.4 Scope of the Project

The scope of this project focuses on developing an intelligent food-label analysis system that accepts both image-based and text-based ingredient input.

The system performs OCR-based ingredient extraction, text preprocessing, deep learning classification, and dietary restriction evaluation. Users can select preferred restriction categories, and the system provides a recommendation along with identified ingredients that influence the decision.

The project focuses on assisting users in making informed food choices and does not replace professional medical advice or clinical diagnosis. Future improvements may include larger datasets, additional dietary categories, multilingual support, and mobile deployment.

Chapter 2

Dataset and Methodology

2.1 Dataset

The dataset used in this project contains food ingredient information with corresponding classification labels. The dataset consists of ingredient descriptions and broad categories such as `Contains` and `Does not contain`. These labels represent whether specific food components are present in a product.

The dataset was mainly used for training and evaluating deep learning classification models. Since the available dataset does not provide individual labels for every dietary category such as lactose, gluten, nuts, vegan, or sugar sensitivity, a hybrid approach was adopted. The trained deep learning model performs ingredient classification, while an explainable rule-based module performs detailed dietary restriction matching.

The dataset used in this project was collected from Kaggle, a publicly available platform for machine learning datasets. The dataset contains food ingredient information with corresponding classification labels such as `Contains` and `Does not contain`. The dataset was selected because it provides suitable ingredient-based information for developing and evaluating deep learning classification models.

2.2 Data Preprocessing

Data preprocessing is an essential step because raw ingredient information may contain inconsistent formatting, unnecessary characters, and variations in ingredient descriptions.

The preprocessing pipeline consists of the following steps:

- **Text Cleaning:** Unnecessary symbols, punctuation, and extra spaces are removed from ingredient descriptions.

- **Normalization:** Ingredient names are converted into a consistent format to reduce variations between similar terms.
- **Tokenization:** Ingredient sentences are divided into meaningful tokens suitable for deep learning input.
- **Feature Preparation:** The processed ingredient text is converted into numerical representations required by neural network models.

For image-based input, additional preprocessing is applied before OCR extraction. OpenCV techniques are used to improve image quality by reducing noise, adjusting contrast, and preparing the image for text recognition.

2.3 Image Processing and OCR

The LabelLens system supports food-label image input. Since images cannot be directly processed by the classification model, an OCR pipeline is applied to extract ingredient information.

The image processing workflow consists of:

- Image acquisition from user upload.
- Image enhancement using OpenCV preprocessing techniques.
- Text extraction using EasyOCR.
- Cleaning and normalization of extracted ingredient text.
- Passing the processed text to the deep learning classification model.

The OCR module allows users to analyze real-world food labels without manually typing ingredient information.

2.4 Deep Learning Model Development

To determine the most suitable classification model for the LabelLens system, three deep learning architectures were developed and evaluated:

- Long Short-Term Memory Network (LSTM)
- Convolutional Neural Network (CNN)
- Deep Neural Network (DNN)

All models were trained using the same preprocessed dataset and evaluated using classification accuracy, precision, recall, F1-score, and validation loss.

2.4.1 Long Short-Term Memory (LSTM) Model

LSTM is a type of recurrent neural network designed to process sequential data. Since ingredient descriptions contain sequential word relationships, LSTM was considered as a possible solution for ingredient classification.

The LSTM model uses memory cells and gates to capture dependencies between ingredient tokens. However, during evaluation, the model showed weaker classification performance compared to other architectures.

2.4.2 Convolutional Neural Network (CNN) Model

CNN models are commonly used for extracting local patterns from structured data. In this project, the CNN architecture was applied to processed ingredient representations to identify important feature patterns associated with food categories.

The CNN model achieved strong classification performance and demonstrated the ability to identify important ingredient patterns.

2.4.3 Deep Neural Network (DNN) Model

A Deep Neural Network consists of multiple fully connected layers that learn complex relationships between input features and output classes.

The DNN model was developed using multiple dense layers with activation functions and optimization techniques suitable for ingredient classification. Because ingredient information was converted into structured numerical features, the DNN architecture was suitable for this classification task.

The DNN model was selected as the final deployment model because it achieved strong classification performance while producing the lowest validation loss among the evaluated models.

2.5 Training Configuration

The deep learning models were trained using the prepared dataset after preprocessing. The training process included data splitting, model optimization, and performance evaluation.

The models were evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Validation Loss

Accuracy measures the overall percentage of correct predictions, while validation loss indicates how effectively the model minimizes prediction error on unseen data.

2.6 Model Performance Comparison

The performance of the trained models was compared to identify the most suitable architecture for deployment.

Table 2.1: Performance Comparison of Deep Learning Models

Model	Accuracy	Precision	Recall	F1-score	Selection
LSTM	72.50%	0.53	0.72	0.61	Not Selected
CNN	98.75%	0.99	0.99	0.99	Not Selected
DNN	98.75%	0.99	0.99	0.99	Selected

The LSTM model achieved an accuracy of 72.50%, showing limited performance for this dataset. CNN and DNN achieved significantly better results with an accuracy of 98.75%.

Although CNN and DNN achieved similar classification accuracy, the DNN model produced a lower validation loss during training. Lower validation loss indicates better generalization capability and reduced prediction error. Therefore, the DNN model was selected as the final model for deployment in the LabelLens application.

2.7 Validation Accuracy Comparison

The validation accuracy of the three deep learning models was compared during training.

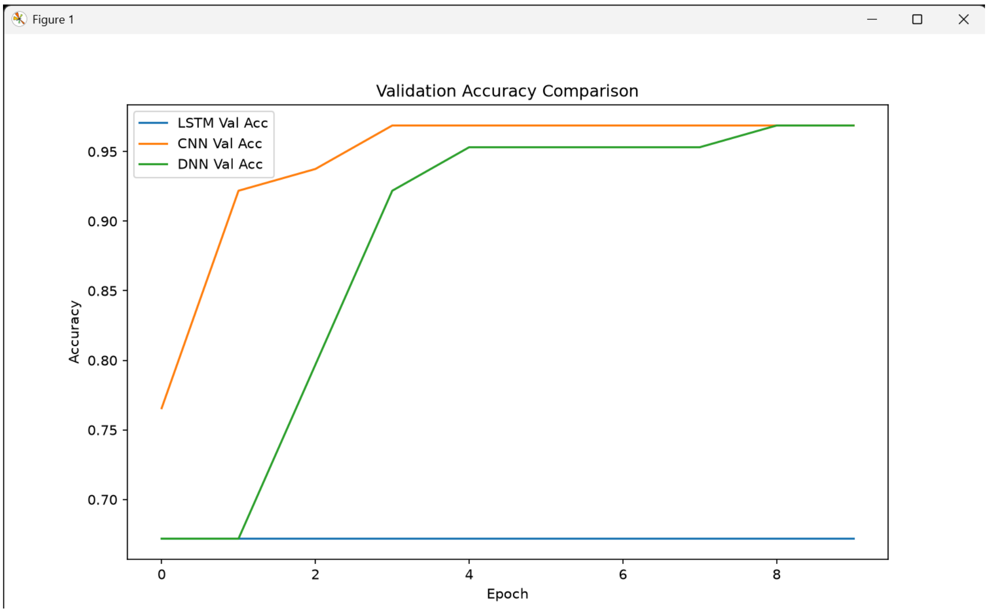

Figure 2.1: Validation Accuracy Comparison of LSTM, CNN, and DNN Models

2.8 System Workflow

The complete workflow of the proposed LabelLens system is:

1. User uploads a food-label image or enters ingredient text manually.
2. Image preprocessing is performed using OpenCV when image input is provided.
3. EasyOCR extracts ingredient information from the image.
4. Extracted text is cleaned and normalized.
5. Processed ingredient information is passed to the trained DNN model.
6. The prediction result is combined with the dietary restriction analysis module.
7. The system identifies risky ingredients and generates a recommendation.

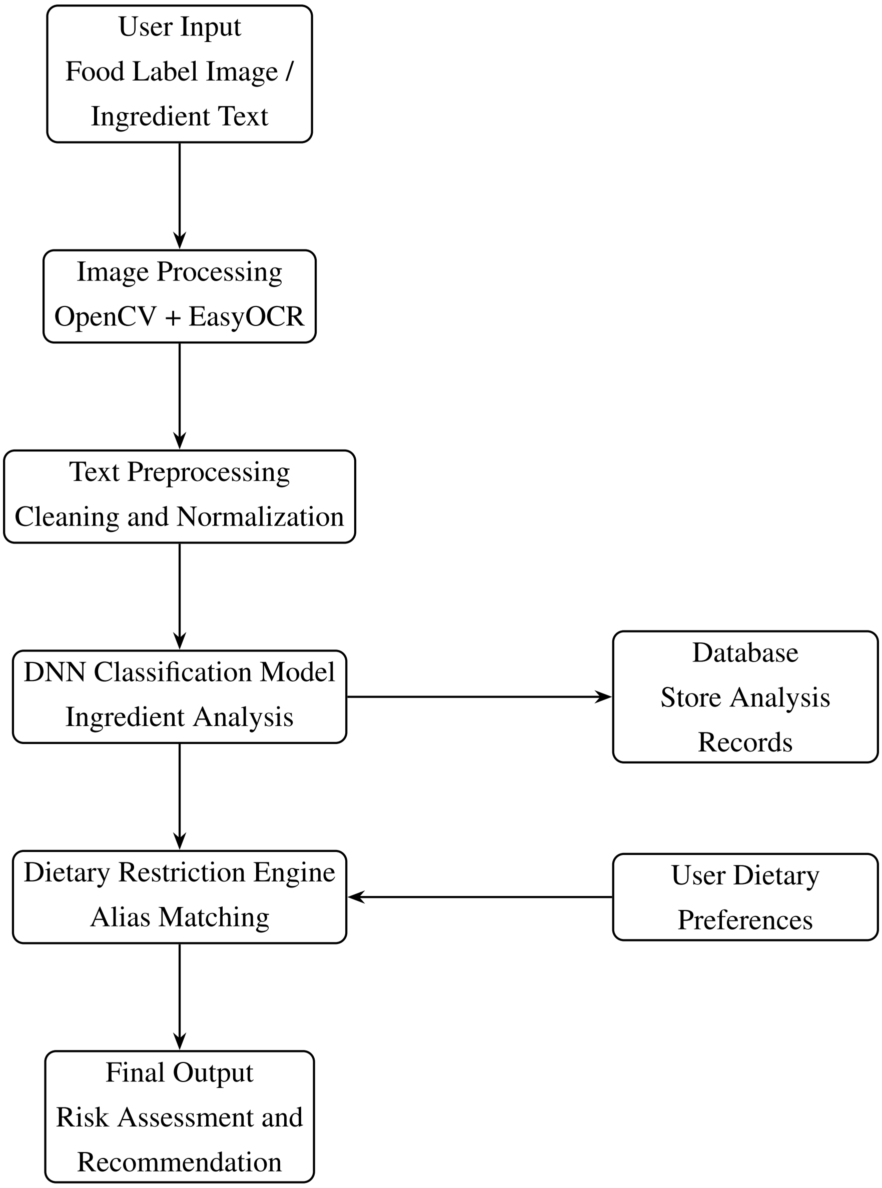

Figure 2.2: Workflow Diagram of the LabelLens Ingredient Analysis System

Chapter 3

System Design and Implementation

3.1 System Architecture

The LabelLens system is designed as a hybrid intelligent application that combines deep learning classification with explainable rule-based dietary analysis. The architecture consists of multiple processing modules that work together to transform raw food-label input into a meaningful dietary recommendation.

The major components of the system are:

- User Interface Module
- Image Processing and OCR Module
- Text Preprocessing Module
- Deep Learning Classification Module
- Dietary Restriction Analysis Module
- Database and Result Management Module

The user can provide input either by uploading a food-label image or entering ingredient text manually. For image input, OpenCV preprocessing and EasyOCR are applied to extract ingredient information. The processed ingredient data is then passed to the selected DNN model for classification. Finally, the prediction result is combined with the restriction analysis engine to generate an explainable recommendation.

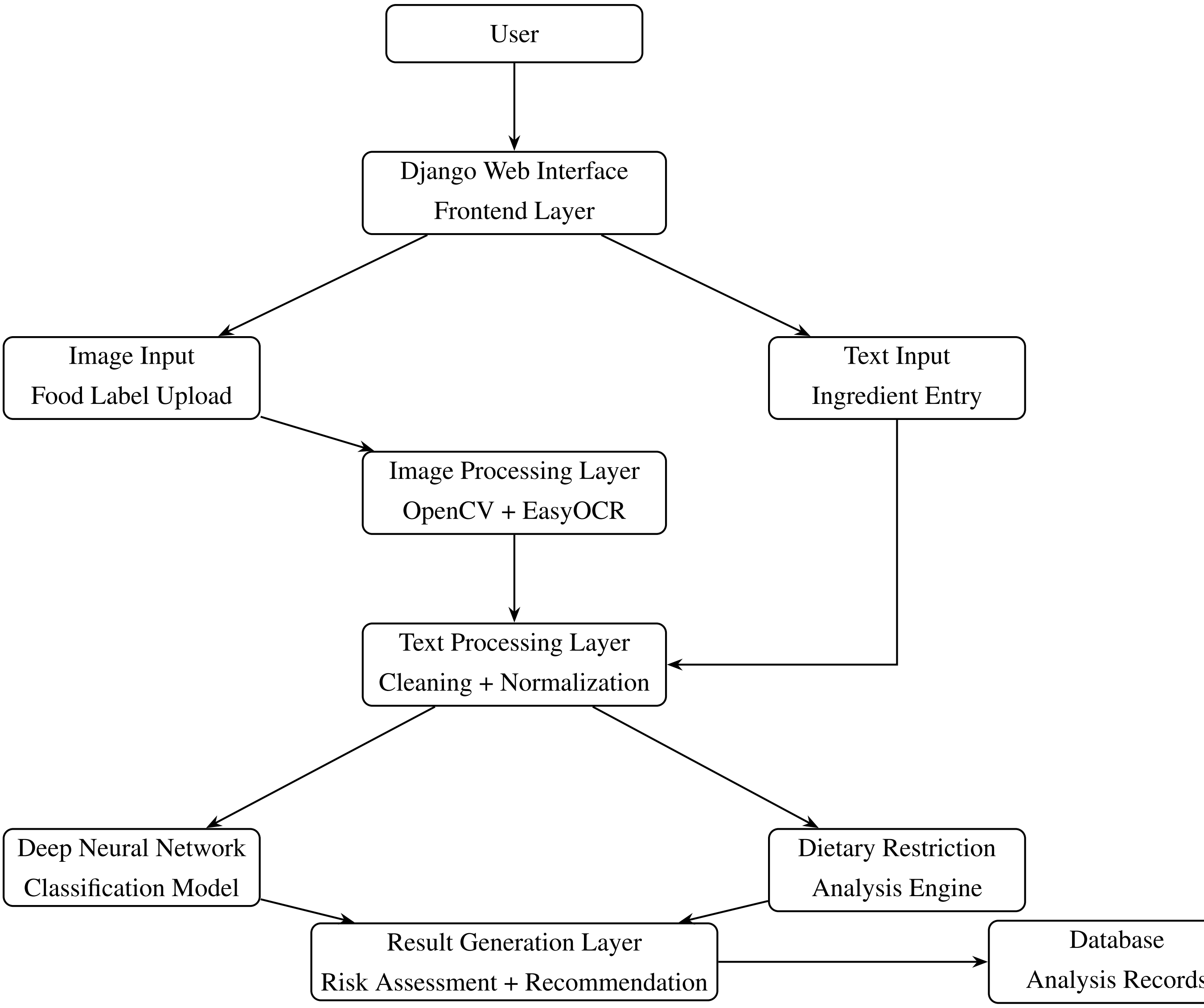

Figure 3.1: System Architecture of the LabelLens Ingredient Analysis System

3.2 Overall System Workflow

The complete processing flow of LabelLens consists of the following stages:

1. **Input Collection:** The user provides either a food-label image or manually entered ingredient text.
2. **Image Processing:** Uploaded images are enhanced using OpenCV techniques to improve OCR performance.
3. **Ingredient Extraction:** EasyOCR extracts ingredient information from the processed image.

4. **Text Processing:** Extracted text is cleaned, normalized, and converted into a suitable format.
5. **Deep Learning Prediction:** The processed ingredient information is passed into the trained DNN model for classification.
6. **Restriction Assessment:** The system compares detected ingredients with predefined dietary restriction rules.
7. **Recommendation Generation:** A final result containing risk status and explanation is displayed to the user.

3.3 Frontend Design

The frontend of LabelLens provides a simple and user-friendly interface that allows users to interact with the system without requiring technical knowledge.

The main features of the frontend include:

- Food-label image upload option.
- Manual ingredient text input.
- Dietary restriction selection options.
- Result display section.
- Explanation of detected ingredients and recommendation.

The interface is designed to make dietary assessment accessible for general users. The system clearly presents the detected risks instead of displaying only a prediction label.

3.3.1 User Interface Screenshots

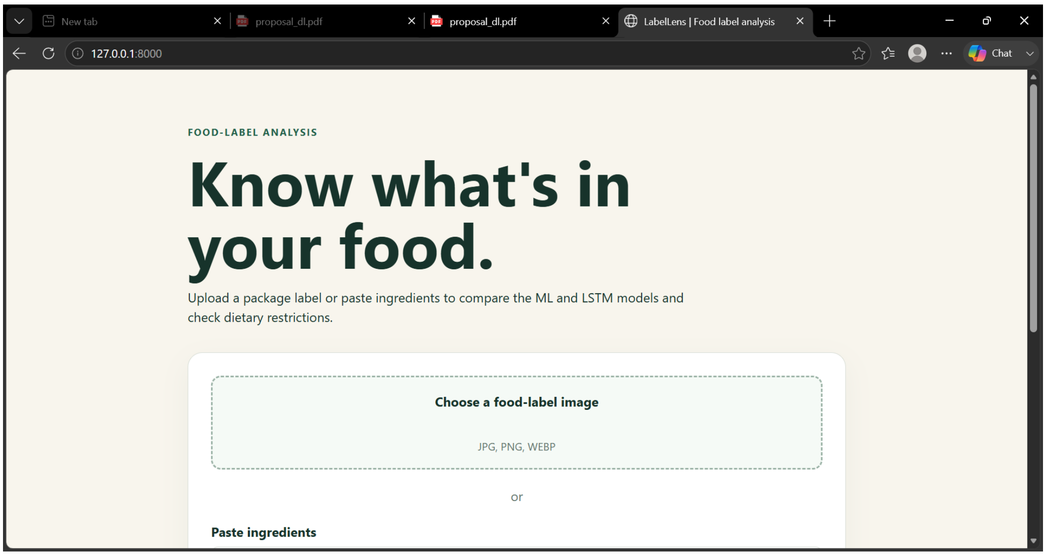

Figure 3.2: LabelLens Home Page Interface

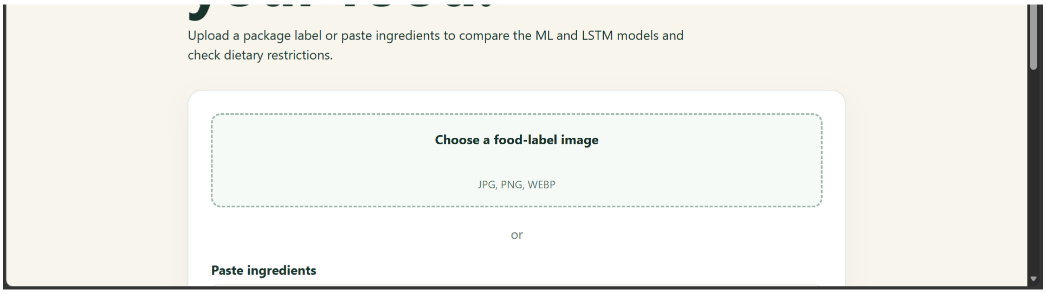

Figure 3.3: Food Label Image Upload Interface

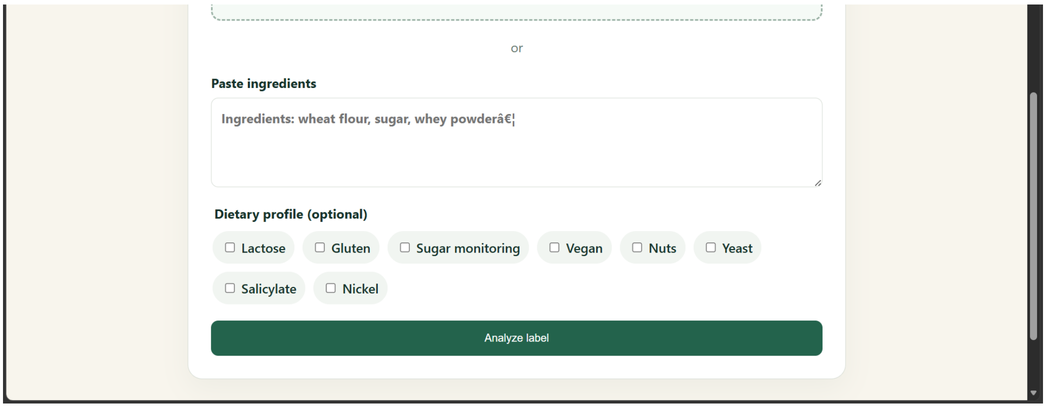

Figure 3.4: Manual Ingredient Text Input Interface

3.4 Backend Design

The backend of the system is developed using Django. It manages user requests, processes uploaded files, communicates with the machine learning model, and returns the final analysis result.

The backend performs the following operations:

- Receiving image or text input from users.
- Applying OCR processing for image-based input.
- Performing ingredient text preprocessing.
- Loading the trained DNN model.
- Generating prediction results.
- Executing dietary restriction matching.
- Returning the final recommendation to the frontend.

The modular backend structure allows future improvements such as adding additional dietary categories, replacing the current model, or integrating larger datasets.

3.5 OCR Processing Module

The OCR module allows the system to analyze real-world food labels captured through images.

The OCR processing pipeline consists of:

1. Image upload from the user.
2. Image enhancement using OpenCV.
3. Text recognition using EasyOCR.
4. Extraction of ingredient information.
5. Text normalization before classification.

OCR performance is highly dependent on image quality. Therefore, preprocessing techniques such as noise reduction and contrast adjustment are applied before text extraction.

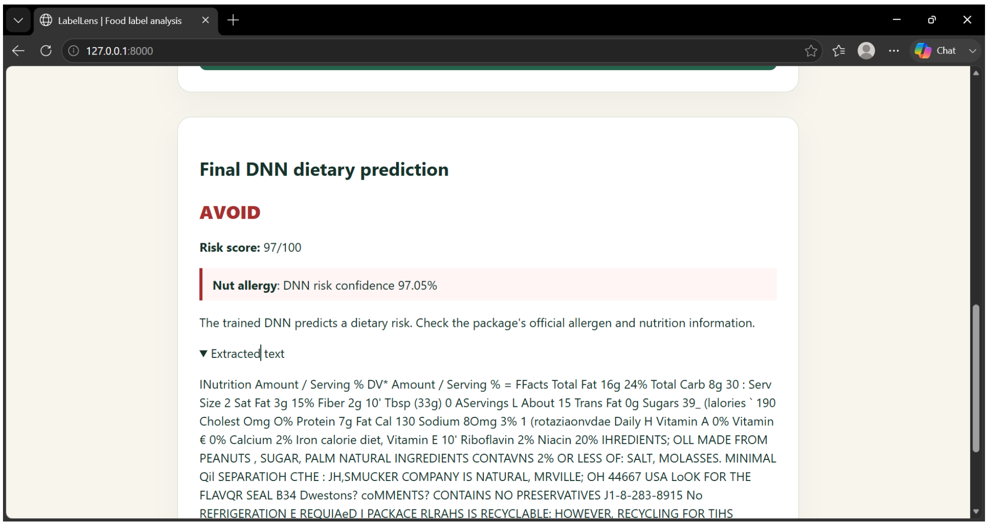

Figure 3.5: Example of OCR Extracted Ingredient Text

3.6 Deep Learning Model Integration

The trained DNN model is integrated into the backend as the main classification component. The integration process consists of:

1. Loading the saved trained model.
2. Receiving processed ingredient features.
3. Performing classification prediction.
4. Returning the predicted category.
5. Combining prediction output with restriction analysis.

The DNN model does not directly replace the explainable restriction engine. Instead, both components work together. The deep learning model provides classification capability, while the rule-based module identifies specific ingredients responsible for dietary warnings.

3.7 Database Design

The application uses a database system to store user interactions and analysis records. The stored information includes:

- Uploaded food-label information.
- Extracted ingredient text.
- Selected dietary restrictions.
- Prediction result.
- Generated recommendation.

Database storage allows users' analysis history to be managed efficiently and supports future expansion of the application.

3.8 Technologies Used

The following technologies were used for implementing LabelLens:

Table 3.1: Technologies Used in the Project

Technology	Purpose
Python	Programming language
Django	Web application framework
TensorFlow/Keras	Deep learning model development
OpenCV	Image preprocessing
EasyOCR	Optical Character Recognition
HTML/CSS/JavaScript	Frontend development
SQLite	Database management

Chapter 4

Results and Discussion

4.1 Experimental Results

The LabelLens system was evaluated from two perspectives:

- Deep learning model performance.
- Application-level functionality.

The deep learning evaluation focused on comparing LSTM, CNN, and DNN models, while application testing evaluated OCR extraction, ingredient analysis, and final dietary recommendation generation.

4.2 Deep Learning Model Evaluation

Three deep learning models were trained and tested using the prepared ingredient dataset. The performance of each model was evaluated using accuracy, precision, recall, F1-score, and validation loss.

Table 4.1: Deep Learning Model Evaluation Results

Model	Accuracy	Precision	Recall	F1-score	Decision
LSTM	72.50%	0.53	0.72	0.61	Not Selected
CNN	98.75%	0.99	0.99	0.99	Not Selected
DNN	98.75%	0.99	0.99	0.99	Selected

The LSTM model achieved an accuracy of 72.50%, which indicates that it was unable to effectively learn the relationship between ingredient features and classification labels.

The CNN model achieved significantly better performance with an accuracy of 98.75%. The model was able to identify important feature patterns from the processed ingredient representations.

The DNN model also achieved an accuracy of 98.75%. However, during training, the DNN produced a lower validation loss compared to the other models. Since lower validation loss indicates better optimization and improved generalization capability, the DNN was selected as the final deployment model.

4.3 Validation Accuracy Comparison

The validation accuracy comparison between the three deep learning models is shown below.

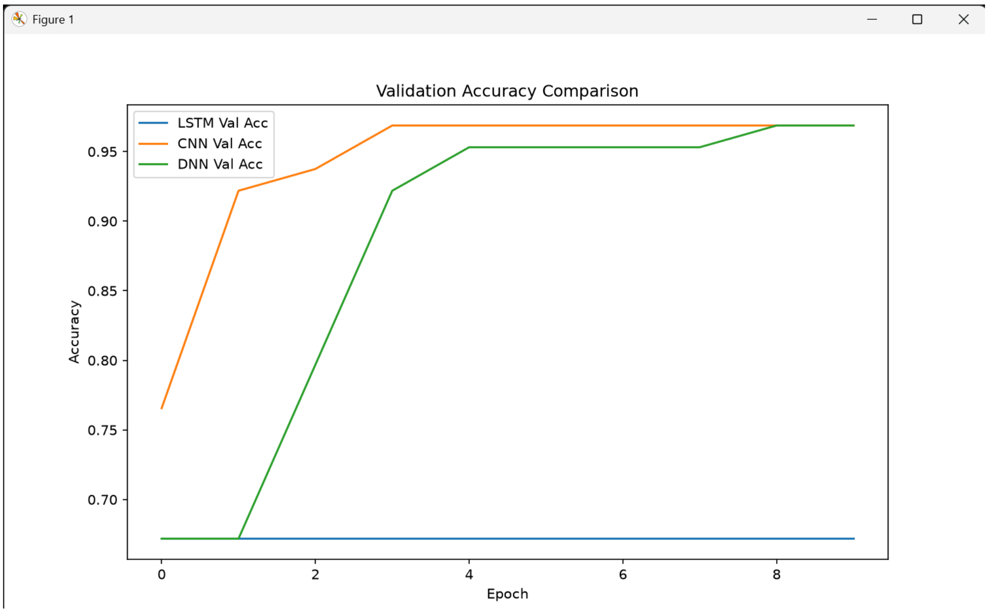

Figure 4.1: Validation Accuracy Comparison of LSTM, CNN, and DNN Models

4.4 Loss Analysis

Loss analysis was considered an important factor during model selection. Although CNN and DNN achieved similar accuracy values, the DNN model demonstrated lower validation loss during training.

A lower validation loss indicates that the model produced fewer prediction errors on unseen data. Therefore, the DNN model was considered more suitable for deployment in the LabelLens application.

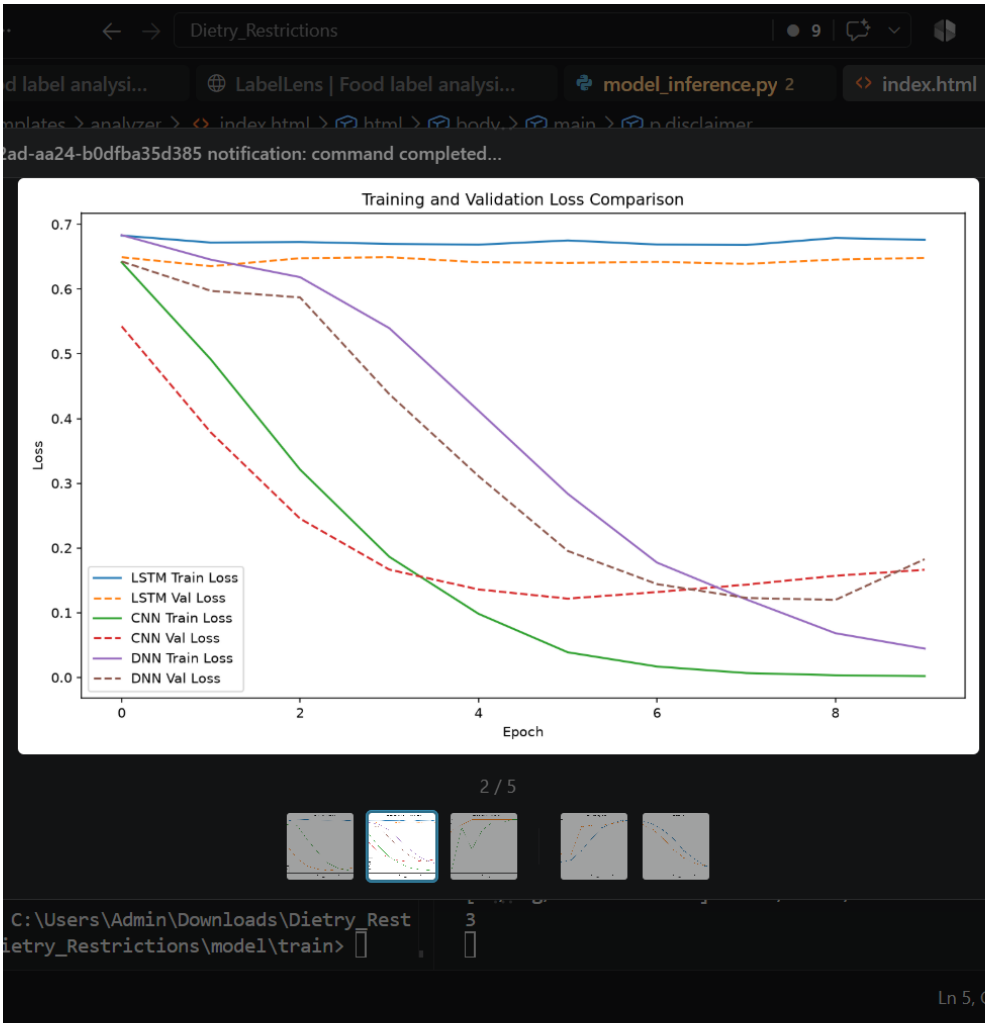

Figure 4.2: Validation Loss Comparison of Deep Learning Models

4.5 Application Testing Results

The developed LabelLens application was tested using both image-based and text-based ingredient inputs.

The following functionalities were successfully tested:

- Uploading food-label images.
- Extracting ingredient information using OCR.
- Processing manually entered ingredient text.
- Performing dietary restriction analysis.
- Generating recommendation messages.
- Displaying results through the web interface.

4.6 Image-Based Ingredient Analysis

For image-based analysis, users upload a food-label image. The system performs preprocessing and OCR extraction before sending the processed ingredient information to the analysis pipeline.

Figure 4.3: Uploaded Food Label Image

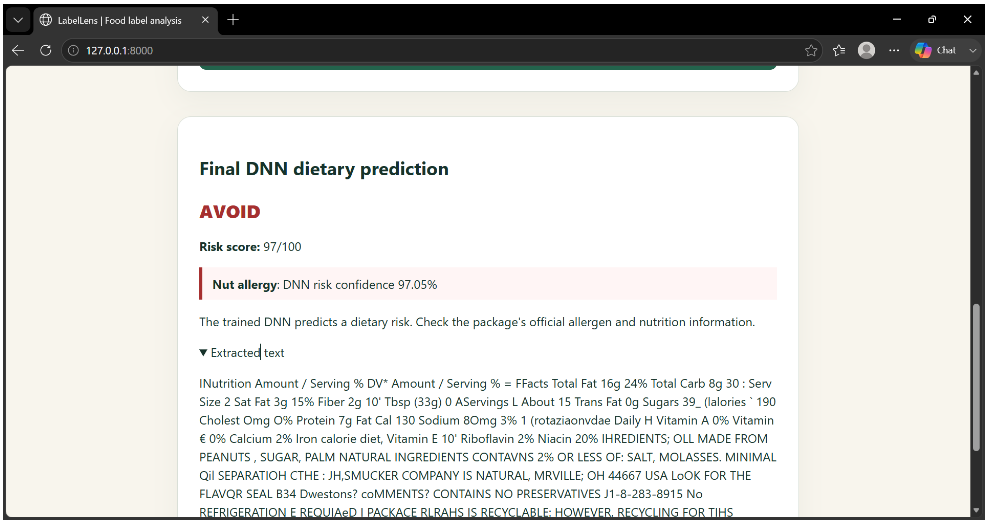

Figure 4.4: Ingredient Text Extracted Through OCR

The extracted text is normalized before being passed into the classification and restriction analysis modules.

4.7 Text-Based Ingredient Analysis

The system also supports direct ingredient input without requiring an image.

Users can manually provide ingredient information, which is then processed by the same normalization and classification pipeline.

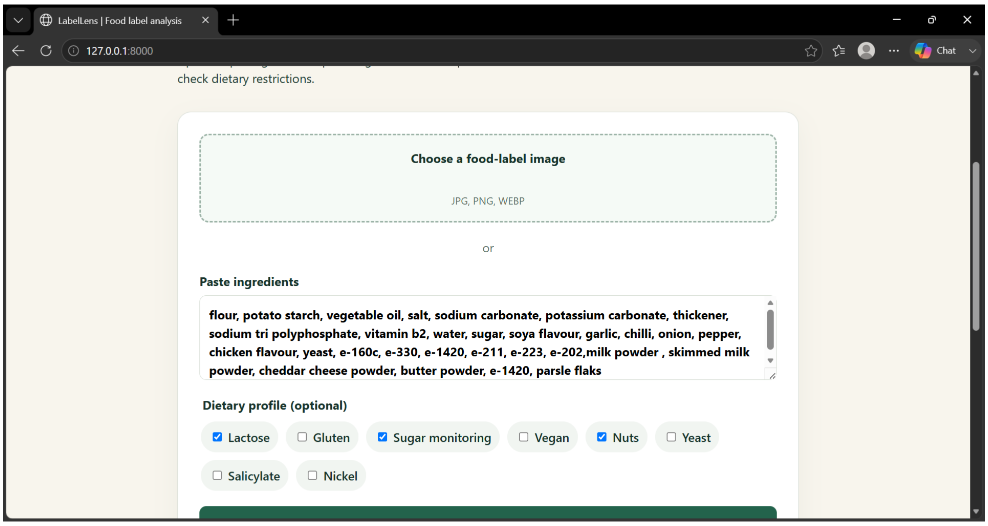

Figure 4.5: Manual Ingredient Text Input

4.8 Final Dietary Assessment Result

After processing the ingredient information, the system provides a final dietary assessment. The output includes:

- Detected dietary risks.
- Matched ingredients responsible for warnings.
- Recommendation message.
- Overall assessment status.

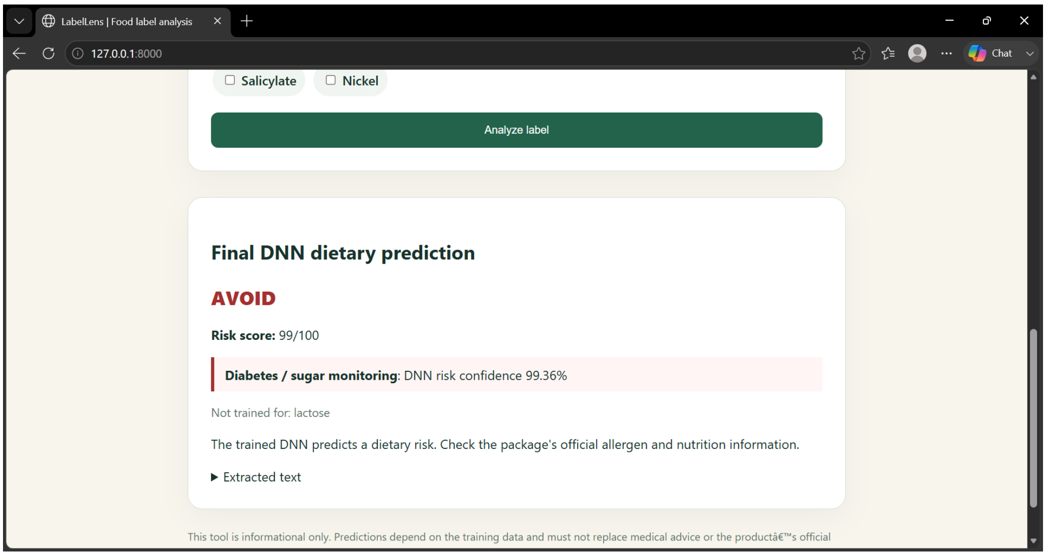

Figure 4.6: Final Dietary Restriction Assessment Result

The explainable output improves user trust because the system does not only provide a classification result but also highlights the ingredient information responsible for the decision.

4.9 Discussion

The experimental results demonstrate that combining deep learning with rule-based analysis provides an effective solution for food-label interpretation.

The deep learning component improves automatic classification capability, while the rule-based module provides transparency by identifying specific ingredients related to dietary restrictions.

A major advantage of LabelLens is explainability. Unlike a fully black-box prediction system, the application informs users about the detected ingredients that influence the recommendation.

However, the system has several limitations. OCR performance may decrease when images contain poor lighting, low resolution, or distorted text. Additionally, the restriction dictionary requires continuous improvement because new ingredient variations may appear in commercial food products.

The current dataset also limits supervised learning because it contains broad labels rather than direct dietary restriction categories. Future improvements can include collecting a larger dataset with detailed restriction-specific annotations.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This project presented **LabelLens**, a deep learning-based ingredient analysis system designed to assist users in evaluating food products according to dietary restrictions.

The proposed system combines multiple artificial intelligence techniques including image preprocessing, Optical Character Recognition (OCR), natural language processing, deep learning classification, and explainable rule-based analysis. The system accepts food-label images or manually entered ingredient information and provides users with an understandable dietary assessment.

Three deep learning models, namely LSTM, CNN, and DNN, were developed and evaluated. Although CNN and DNN achieved similar classification accuracy, the DNN model was selected as the final model because it demonstrated lower validation loss and better suitability for deployment.

The developed Django-based application successfully integrates the trained DNN model with OCR-based ingredient extraction and a restriction analysis module. The system not only provides predictions but also identifies relevant ingredients responsible for dietary warnings, improving transparency and user confidence.

Overall, the project demonstrates that combining deep learning techniques with explainable decision-making methods can provide an effective solution for automated food-label analysis and dietary restriction assessment.

5.2 Future Work

Although the developed system achieves its objectives, several improvements can be implemented in future work.

Possible future improvements include:

- Expanding the dataset with more food products and detailed restriction-specific labels.
- Improving the deep learning model using larger and more diverse training data.
- Enhancing OCR performance through advanced image enhancement techniques.
- Adding multilingual ingredient recognition support.
- Developing a mobile application version for easier user access.
- Integrating a cloud-based deployment system for large-scale usage.
- Adding more dietary categories and continuously updating ingredient databases.

Acknowledgments

The successful completion of this project required continuous learning, technical exploration, and support from various sources.

First, we would like to express our gratitude to our course instructor and supervisor for providing valuable guidance, feedback, and encouragement throughout the development of this project.

We would also like to acknowledge the contribution of open-source technologies and research communities that provided the tools and frameworks used in this work.

Project Contributor

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AI Assistance Tools

The following AI-based tools were used for technical assistance, documentation support, and improving project development:

- **ChatGPT (OpenAI):** Used for project planning, debugging assistance, documentation improvement, and report preparation.
- **Google Gemini:** Used for technical discussion, research support, and content refinement.

Technologies Used

The following technologies were used during the development of LabelLens:

- Python
- Django
- TensorFlow / Keras
- OpenCV
- EasyOCR
- HTML
- CSS
- JavaScript
- SQLite Database

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